Math 1232 Summer 2025
Mastery Quiz 9
Due Thursday, July 31

This mastery quiz has three topics, and everyone should do all three. Please do your best, but don't worry if you make a minor error—your goal is to demonstrate your mastery of the underlying material.

Feel free to consult your notes, but please don't discuss the actual quiz questions with other students in the course.

Remember that you are trying to demonstrate that you understand the concepts involved. For all these problems, justify your answers and explain how you reached them. Do not just write "yes" or "no" or give a single number.

Please turn in this quiz in class on Tuesday. You may print this document out and write on it, or you may submit your work on separate paper; in either case make sure your name is clearly on it. If you absolutely cannot turn it in in person, you can submit it electronically but this should be a last resort.

Topics on this quiz:

- Major Topic 3: Series Convergence
- Major Topic 4: Taylor Series
- Secondary Topic 7: Power Series

Name: Solutions

Name: Solutions

Major Topic 3: Series Convergence

Remember to show *all* your work—this includes stating any convergence tests you use, as well as computing any limits that arise.

(a) Analyze the convergence of the series $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n 3^n}{5^n+1}$.

Solution. We use the Ratio test. We have

$$\begin{aligned}
\lim_{n\to\infty} \left| \frac{\frac{(-1)^{n+1}3^{n+1}}{5^{n+1}+1}}{\frac{(-1)^n 3^n}{5^n+1}} \right| &= \lim_{n\to\infty} \frac{3^{n+1}(5^n+1)}{3^n(5^{n+1}+1)} \\
&= \lim_{n\to\infty} 3\frac{5^n+1}{5^{n+1}+1} \\
&= \lim_{n\to\infty} 3\frac{1+1/5^n}{5+1/5^n} = \frac{3}{5}.
\end{aligned}$$

This limit is less than 1, so by the ratio test this converges absolutely.

(b) Analyze the convergence of the series $\displaystyle\sum_{n=1}^{\infty} \frac{n\sin(n)}{n^3+2}$.

Solution. This series has positive and negative terms, but it's not alternating. We basically have to look at absolute convergence.

We consider the series

$$\sum_{n=1}^{\infty} \left| \frac{n\sin(n)}{n^3+2} \right| = \sum_{n=1}^{\infty} \frac{n|\sin(n)|}{n^3+2}.$$

We can't really use the limit comparison test here, because the $\sin(n)$ will screw it up. But we can use the usual comparison test. We know that $0 \leq |\sin(n)| \leq 1$, so

$$\frac{n|\sin(n)|}{n^3+2} \leq \frac{n}{n^3} = \frac{1}{n^2}.$$

We know that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges by the p-series test, so this series converges by the comparison test. Thus our original series converges absolutely.

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(c) Analyze the convergence of $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n \sqrt{n}}{2n+3}$.

Solution. This is an alternating series. Since the terms $\frac{\sqrt{n}}{2m+3}$ decrease to zero as n goes to infinity, this converges by the alternating series test.

However, it doesn't absolutely converge. If we look at the absolute value series, we have $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{2n+3}$. You can see this doesn't converge in a couple ways. The integral test isn't super plausible here. You can do a comparison test to $\frac{1}{\sqrt{n}}$: this is larger than $\frac{1}{3\sqrt{n}}$ for large n, and $\frac{1}{3\sqrt{n}}$ diverges. (Note: this is <u>not</u> larger than $\frac{1}{\sqrt{n}}$.)

It may be easier to use the limit comparison test, though. We have

$$\lim_{n\to\infty} \frac{\frac{\sqrt{n}}{2n+3}}{\frac{1}{\sqrt{n}}} = \lim_{n\to\infty} \frac{n}{2n+3} = 1/2.$$

Since the series $\sum \frac{1}{\sqrt{n}}$ diverges, by the limit comparison test, $\sum \frac{\sqrt{n}}{2n+3}$ diverges, and thus our series does not converge absolutely.

Name: Solutions

Major Topic 4: Taylor Series

(a) Write a power series expression for $\frac{2x^2}{4x+1}$ centered at 0. What is the radius of convergence?

Solution. We know that

$$\begin{aligned} \frac{1}{1-(-4x)} &= \sum_{n=0}^{\infty}(-4x)^n \\ \frac{2x^2}{1+4x} &= 2x^2\sum_{n=0}^{\infty}(-4)^n x^n \\ &= \sum_{n=0}^{\infty} 2\cdot(-4)^n x^{n+2} \\ \text{(or)} \qquad &= \sum_{n=2}^{\infty} 2^{2n-3}(-1)^n x^n. \end{aligned}$$

The radius of convergence is $1/4$. We can figure that out by reasoning from the geometric series: the radius of convergence for the geometric series is 1, so it converges for $-1 < -4x < 1$ or $-1/4 < x < 1/4$. Or we can use the ratio test:

$$\lim_{n\to\infty}\left|\frac{2^{2n-1}(-1)^{n+1}x^{n+1}}{2^{2n-3}(-1)^n x^n}\right| = \lim_{n\to\infty} 4|x|$$

and thus it converges when $4|x| < 1$.

(b) If $f(x) = \sum_{n=0}^{\infty}\frac{n+1}{n!+1}x^n$, compute $\int_3^6 f(x)$.

Solution.

$$\begin{aligned} \int f(x) &= \sum_{n=0}^{\infty}\frac{1}{n!+1}x^{n+1} + C \\ \int_3^6 f(x) &= \sum_{n=0}^{\infty}\frac{1}{n!+1}(6^{n+1}-3^{n+1}). \end{aligned}$$

Name: Solutions

(c) Write a power series expression for $\ln(1+x^2)$ centered at 0. What is the radius of convergence?

Solution. There are a few ways to approach this. One is to observe that $\frac{d}{dx}\ln(1+x^2) = \frac{2x}{1+x^2}$. We know that

$$\frac{1}{1+x^2} = \frac{1}{1-(-x^2)} = \sum_{n=0}^{\infty}(-x^2)^n = \sum_{n=0}^{\infty}(-1)^n x^{2n}$$
$$\frac{2x}{1+x^2} = 2x\sum_{n=0}(-1)^n x^{2n} = \sum_{n=0}^{\infty}(-1)^n 2x^{2n+1}$$

Then we can integrate both sides:

$$\begin{aligned}\ln(1+x^2) &= \int \sum_{n=0}^{\infty} 2(-1)^n (x)^{2n+1}\,dx \\ &= C + \sum_{n=0}^{\infty} 2(-1)^n \frac{x^{2n+2}}{2n+2}\end{aligned}$$

and plugging 0 in on both sides tells us that $C = 0$. So our power series is

$$\ln(1+x^2) = \sum_{n=0}^{\infty} 2(-1)^n \frac{x^{2n+2}}{2n+2}.$$

Alternatively, you could recall that

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n$$

so we can compute

$$\begin{aligned}\ln(1+x^2) &= \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n \\ &= \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^{2n}.\end{aligned}$$

This looks different from our previous answer because it's been reindexed, but it is in fact the same answer. As an exercise, see if you can show they're the same!

Name: Solutions

Secondary Topic 7: Power Series

(a) Find the radius of convergence and the interval of convergence of $\displaystyle\sum_{n=1}^{\infty} \frac{(2x-5)^n}{n^2}$.

Solution. We use the ratio test.

$$\lim_{n\to\infty} \left| \frac{\frac{(2x-5)^{n+1}}{(n+1)^2}}{\frac{(2x-5)^n}{n^2}} \right| = \lim_{n\to\infty} \left| \frac{(2x-5)n^2}{(n+1)^2} \right|$$

$$= |2x-5| \lim_{n\to\infty} \frac{n^2}{(n+1)^2} = |2x-5|.$$

So we need $|2x-5| < 1$ or $-1 < 2x-5 < 1$, or $4 < 2x < 6$ or $2 < x < 3$. So the radius is $1/2$.

To find the interval we need to check the endpoints. We see

$$\sum_{n=0}^{\infty} \frac{(4-5)^n}{n^2} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n^2} \quad \text{converges by alternating series test}$$

$$\sum_{n=0}^{\infty} \frac{(6-5)^n}{n^2} = \sum_{n=0}^{\infty} \frac{1}{n^2} \quad \text{converges by } p\text{-series test}$$

(b) Find the radius of convergence and the interval of convergence of $\displaystyle\sum_{n=1}^{\infty} \frac{n^2 x^n}{(2n)!}$.

Solution. We use the ratio test.

$$\lim_{n\to\infty} \left| \frac{\frac{(n+1)^2 x^{n+1}}{(2n+2)!}}{\frac{n^2 x^n}{(2n)!}} \right| = \lim_{n\to\infty} \left| \frac{(n+1)^2 x}{n^2(2n+2)(2n+1)} \right|$$

$$= |x| \lim_{n\to\infty} \frac{(n+1)^2}{n^2(2n+2)(2n+1)} = 0.$$

This is always less than 1, so the series always converges. The radius of convergence is ∞ and the interval of convergence is $(-\infty, \infty)$.